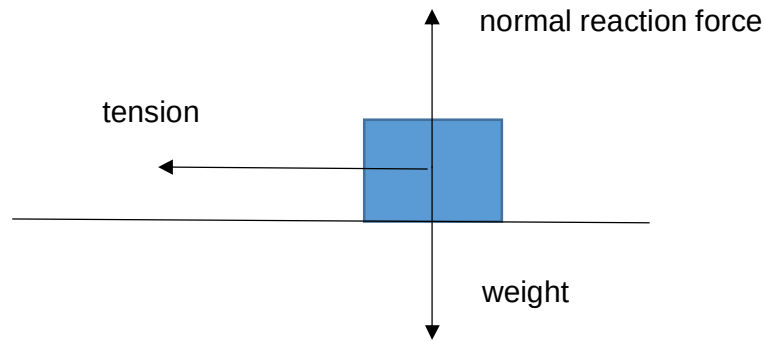


4 (a) (i)



$$ma = -kx$$

$$a = -(k/m)x$$

Since the resultant force on P is the elastic force by the spring, the acceleration on P is directly proportional to the displacement of P from the equilibrium position.

Or **tension is directly proportional to displacement** from equilibrium position and is the resultant force (Second B1 mark)

The negative sign also shows that the direction of acceleration is always directed towards the equilibrium position.

(ii)

$$\omega^2 = \frac{k}{m}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$T = 2\pi \sqrt{\frac{5}{100}}$$

$$= 1.405 \text{ s}$$

$$= 1.4 \text{ (2 s.f.)}$$

(b) (i) Loss in elastic potential energy = Gain in kinetic energy

$$\frac{1}{2}kx^2 = \frac{1}{2}(M + m)v^2$$

$$\frac{1}{2}(100)(0.2)^2 = \frac{1}{2}(5.0 + 3.0)v^2$$

$$v = 0.707 \text{ m s}^{-1}$$

(ii) Gain in elastic potential energy = Loss in kinetic energy

$$\begin{aligned}\frac{1}{2}kx^2 &= \frac{1}{2}mv^2 \\ \frac{1}{2}(100)(x)^2 &= \frac{1}{2}(5.0)(0.707)^2 \\ x &= 0.158 \text{ m}\end{aligned}$$

OR

$$\begin{aligned}v &= \omega x_0 \\ 0.7071 &= \frac{2\pi}{1.405}x_0 \\ x_0 &= 0.158 \text{ m}\end{aligned}$$

(iii) Distance travelled by Q in quarter of a period
 $= 0.707 \times 0.25(1.405) = 0.2484 \text{ m}$

Distance apart $= 0.2484 - 0.1581 = 0.0903 \text{ m} = 0.090 \text{ m}$ (2 s.f.)

(c) The frequency will be lower since period will be longer according to

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Amplitude will be same since maximum elastic potential energy of system remains the same.